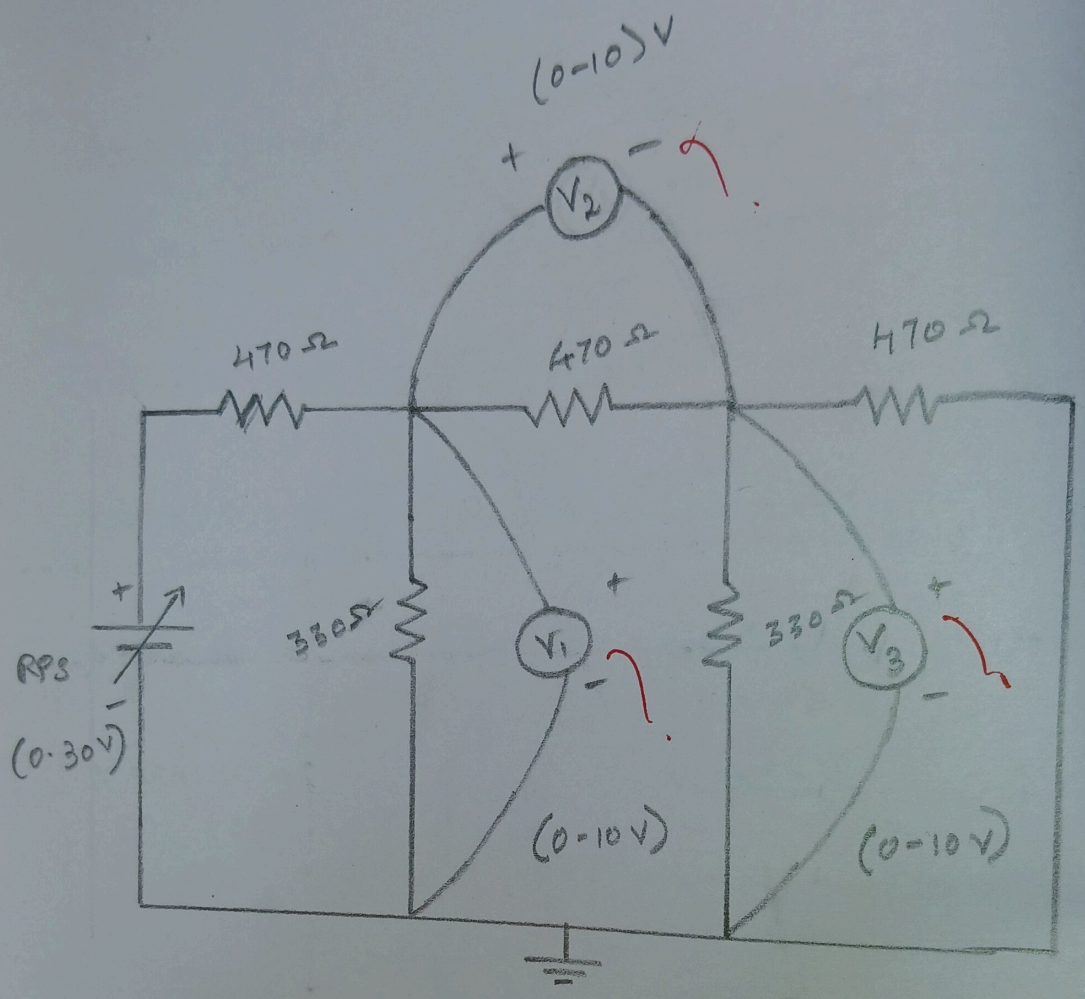
(0-10)V
+
-
V2
470 Ω
470 Ω
470 Ω
+
RPS
-
(0-30V)
330Ω
+
V1
-
(0-10 V)
330Ω
+
V3
-
(0-10 V)

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DETERMINATION OF VOLTAGE USING NODAL ANALYSIS

AIM:

To determine the ~~th~~ voltage in the circuit using the nodal analysis both theoretically and practically for the DC circuit.

Apparatus required:

SL. NO:	APPARATUS	SPECIFICATION	QTY
1.	Regulated power Supply	(0-30 V)	1
2.	Multimeter	–	1
3.	Resistors	470Ω, 330Ω	3,2
4.	Bread board	–	1

Theory:

In electric circuits analysis, nodal analysis, node voltage analysis or the branch current method is determining the voltage (potential difference) b/w nodes in electric circuits in term of branch current.

PROCEDURE:

1. Give the connections as per diagram
2. Switch ON The supply. Vary the RPS. and set a particular input voltage.
3. Note the reading of the ammeter and voltmeters and tabulate them.
4. Vary the RPS of different input of voltage and note all the readings
5. Reduce RPS to its minimum and OFF the supply.
6. With the noted values verify the Kirchoff's laws partically and verify it theoretically.

Parameters	Theoretical	Practical
V_A	0.76	
V_B	0.61	
$V_A - V_B$	0.15	

S.No	Description	Marks Allotted	obtained
1.	Setup / connection	20	20
2.	observation / Readings	20	18
3.	~~observation / Result~~	~~20~~	18
4.	calculation	20	18
5.	viva	20	18
6.	Record	20	20
			94

RESULT:

Nodal analysis is verified practically and theoretically

a) The voltage V_1 is (V) 0.69 V

b) The voltage V_2 is (V) 0.49 V

c) The voltage V_3 is (V) 0.9 V